

SPML Summer 2023

Generative AI

July 6, 2025

1 Signal Processing and Machine Learning (SPML) – Summer 2023

1.1 Generative AI

July 6, 2025

2 Overview of the Course

The course has evolved from an introduction to basic machine learning (ML) concepts to covering Generative Modeling and Generative AI within the context of Signal Processing (SP).

2.1 Tentative Outline

- Part I – Estimation and Regression
- Part II – Generative Modeling
- Part III – Applications

3 Part I – Estimation and Regression

3.1 Introduction

What is the difference between estimation and regression? In both cases, we want to find the relationship between variables. Estimation is solely based on observations of random variables, whereas regression requires observations of pairs consisting of dependent and independent random variables.

3.2 Estimation

Suppose we have an estimator:

$$\hat{y} = 2 \tag{1}$$

Statistical models are represented explicitly via data samples:

$$(x_i, y_i) \in S, \quad |S| = s \tag{2}$$

where the observed variable $x \in \mathbb{R}$ has the distribution

$$x_i \sim f(x_i) = \int f_{X,Y}(x_i, y) dy \tag{3}$$

if a statistical model exists. The estimated variable $y \in \mathbb{R}^m$ is characterized by the conditional distribution:

$$y_i \sim f_{Y|X}(y_i|x) \tag{4}$$

3.3 Model Assumptions and Estimation

In cases where a statistical model is partially or fully available, the inference process (handled by the SSP team) aims to produce optimal solutions.

3.3.1 Maximum Likelihood (ML) Estimation

When partial knowledge of the joint distribution $f_{X,Y}(x,y)$ exists, the maximum likelihood estimator (MLE) is defined as:

$$\hat{\text{ML}} = \arg \max_{y \in \mathcal{Y}} \sum_{i=1}^n \log f_{Y|X}(y_i|x) \quad (5)$$

assuming the observations $Q = \{x_1, x_2, \dots, x_q\}$ are independent and identically distributed (i.i.d.) and originate from the true, unknown y .

3.4 Further Assumptions and Asymptotic Properties

- Under the assumption that $x_i \sim f_x(x_i)$, fixed-condition ML estimators are asymptotically efficient.
- As the number of samples $n \rightarrow \infty$, the estimator $\hat{\text{ML}}$ approaches the minimum achievable variance, the Cramér-Rao bound.

$$\boxed{(6)} \quad (6)$$

3.5 Conditional Mean Estimation

The conditional mean estimate Y_{CME} is given by:

$$Y_{CME} = \int y f_{Y|X_1, \dots, X_q}(y|x_1, \dots, x_q) dx dy \quad (7)$$

which can also be expressed as:

$$Y_{CME} = \mathbb{E}_{Y|X_1, \dots, X_q}[y \mid X_1 = x_1, \dots, X_q = x_q] \quad (8)$$

3.6 Conditional Mean Estimator

The conditional mean estimator (CME) of Y given $X = x$ is:

$$\hat{Y}_{CE} = \mathbb{E}_Y[y|X = x] = \int y f_{Y|X}(y|x) dy \quad (9)$$

with the conditional density proportional to the joint:

$$f_{Y|X}(y|x) \propto f_{XY}(x, y) \quad (10)$$

where $f_Y(y)$ is the prior distribution of Y .

3.7 Optimality of the Conditional Mean Estimator

The CME minimizes the mean squared error (MSE):

$$\boxed{\hat{Y}_{CE} = \arg \min_Y \mathbb{E}_Y [(Y - \hat{Y})^2]} \quad (11)$$

making it the optimal predictor in the MSE sense.

4 Linear Minimum MSE Estimator

Suppose the following covariances:

$$C_x = \text{Var}[X] = \mathbb{E}_x [(X - \mu_x)^2], \quad (12)$$

$$C_y = \text{Var}[Y] = \mathbb{E}_y [(Y - \mu_y)^2], \quad (13)$$

$$C_{xy} = \text{Cov}[X, Y] = \mathbb{E}_{x,y} [(X - \mu_x)(Y - \mu_y)], \quad (14)$$

with symmetry $C_{xy} = C_{yx}$.

The linear minimum mean square error (LMMSE) estimator for Y given $X = x$ is:

$$\hat{Y}_{LMMSE} = C_{yx}C_x^{-1}(x - \mu_x) + \mu_y \quad (15)$$

This estimator yields the linear predictor with minimal mean squared error, and the trace of the minimal MSE is:

$$\text{trace} [\text{Cov}_Y - C_{yx}C_x^{-1}C_{xy}] \quad (16)$$

5 Conditional Priors

In many applications, the prior distribution $f_Y(y)$ is known and can be computed as:

$$f_Y(y) = \int f_{Y|x}(y|x)f_x(x) dx \quad (17)$$

where $f_x(x)$ is known, and $f_{Y|x}(y|x)$ is the conditional prior.

The expected value conditioned on $Y = y$ can then be written as:

$$\mathbb{E}_{X|Y=y}[Y|X] = \mathbb{E}_{X|Y=y}[Y|X, f_{Y|x} = f_{Y|x}] \quad (18)$$

6 Proof Sketch

The conditional expectation $\mathbb{E}_{Y|x=z}[Y|X = x]$ can be derived from the joint distribution:

$$\mathbb{E}_{Y|x=z}[Y|X = x] = \int \int_{x,x} x f_{Y|x}(x, y) dx dy \quad (19)$$

$$= \int \int_x x f_{Y|x}(x|y) dx dy \quad (20)$$

$$= \int \left(x f_{Y|x}(x|y) dx \right) f_{Y|x}(y) dy \quad (21)$$

$$= \int f_{x|y}(x|y) dx f_Y(y) dy \quad (22)$$

$$= \int_y f_Y(y) dy \quad (23)$$

This illustrates the relationship between joint and marginal distributions in the context of conditional expectations.

7 Conditional Prediction

The conditional point estimate (CPE) based on the prior is:

$$Y_{CPE} = \mathbb{E}[d_x] \left[\sum_y y f_{x,y}(x|y_{x,y}) dy \right] \quad (24)$$

This essentially represents an average over linear estimators, weighted by the conditional distributions.